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Top Skills

Supabase
Celery
Redis

Bhanu Pratap Rana

Applied AI Engineer | GenAI • Computer Vision • RAG • FastAPI • AI Automation | Building Production-Grade AI Systems
Agra, Uttar Pradesh, India

Summary

Applied AI Engineer specializing in Generative AI, Computer Vision, RAG systems, AI Automation, and scalable AI platform development. I design and build production-grade AI systems using Python, FastAPI, OpenCV, PyTorch, PostgreSQL, Redis, Docker, and modern AI infrastructure. My experience spans AI-powered automation platforms, OCR/document intelligence systems, semantic search engines, multilingual AI applications, and real-time AI-driven products. Currently working at Kendriya Hindi Sansthan as an AI/ML Engineer, contributing to AI-enabled educational and digital empowerment initiatives including ATAL AI. I have also worked with startups on AI automation products, intelligent document systems, real-time voice AI assistants, and full-stack AI applications.

Key Areas of Expertise:

- Generative AI & LLM Applications
- RAG & Vector Search Systems
- Computer Vision & OCR
- FastAPI & Scalable Backend Architecture
- AI Automation Workflows
- Real-time AI Systems
- Full-Stack AI Product Development
- Multilingual & Offline-First Systems

Featured Projects:

- Aura AI — AI-powered makeup recommendation & virtual try-on platform
- ATAL AI — AI-enabled digital literacy and adaptive learning platform
- Hindi Semantic Search System — multilingual RAG-powered search engine for Hindi literature
- Voice AI Assistant — real-time speech-to-text and document-grounded AI assistant
- AI-Powered API Test Case Generator using Groq + ChromaDB
- Smart HR Tool — AI-powered HR automation platform

Passionate about building impactful AI systems that solve real-world problems through intelligent automation, scalable architectures, and user-focused product design. Open to remote AI engineering opportunities, startup collaborations, AI product development, and innovative GenAI projects.

Experience

Kendriya Hindi Sansthan

Research Assistant

March 2025 - Present (1 year 4 months)

Agra

As an Research Assistant at Kendriya Hindi Sansthan, I contribute to AI-driven educational technology, digital empowerment initiatives, and research projects focused on improving learning outcomes in rural India.

Key Contributions:

- Designed and developed the ATAL AI platform, a Progressive Web Application (PWA) supporting digital literacy, adaptive learning, and AI-enabled educational initiatives.
- Built a multilingual Semantic Search Library leveraging Retrieval-Augmented Generation (RAG), vector embeddings, and large language models to enable intelligent knowledge discovery and document retrieval.
- Developed AI-powered assessment and evaluation frameworks supporting personalized learning experiences and educational analytics.
- Conducted large-scale field studies and data collection across Assam, covering 6 government schools and 570+ students in Sualkuchi, Bongshar, Bamundi, Dadra, Amingaon, and Hajo to evaluate digital learning adoption and technology readiness.
- Analyzed educational datasets and prepared research reports to support policy recommendations, AI adoption strategies, and digital learning interventions.
- Contributed to the review and evaluation of 22 research papers and participated in the development of an edited academic book related to AI-enabled education and digital empowerment.
- Designed data pipelines, dashboards, and reporting frameworks using AI, analytics, and visualization tools to support evidence-based decision making.

Impact:

- Achieved student adoption rates of approximately 50–65% during pilot implementations and comparative studies against existing digital learning platforms.
- Supported AI-driven educational initiatives aimed at improving digital literacy and technology accessibility in rural communities.

Technologies:

Python, FastAPI, PostgreSQL, Supabase, OpenAI, RAG, Semantic Search, Power BI, AI/ML, Data Analytics, Progressive Web Applications (PWA), Vector Databases, Generative AI.

Aura AI

Founding AI Engineer

December 2025 - Present (7 months)

India

Leading the design and development of Aura AI, a production-grade AI-powered beauty technology platform providing personalized makeup recommendations, facial analysis, and real-time virtual try-on experiences.

Key Contributions:

- Designed scalable AI architecture using FastAPI, PostgreSQL, Redis, Celery, Docker, and React Native.
- Developed computer vision pipelines using MediaPipe and OpenCV for facial analysis, skin tone detection, and recommendation systems.
- Integrated GPT-4o-powered AI beauty assistant for personalized consultations and intelligent recommendations.
- Implemented asynchronous AI processing pipelines and containerized deployment workflows.
- Developed multilingual mobile experiences supporting international users and scalable product growth.

Technologies:

Python, FastAPI, OpenCV, MediaPipe, GPT-4o, PostgreSQL, Redis, Celery, Docker, React Native, TypeScript.

Bytical.ai

Machine Learning Intern

July 2024 - December 2025 (1 year 6 months)

India

Worked as a Machine Learning Intern contributing to AI, Computer Vision, and intelligent document processing solutions.

Key Contributions:

- Developed machine learning and computer vision workflows using Python, OpenCV, TensorFlow, and PyTorch.
- Built OCR-based document processing systems for automated information extraction and document digitization.
- Developed NLP pipelines for text analysis, semantic processing, and intelligent search applications.
- Assisted in model training, evaluation, deployment, and optimization for production-oriented AI solutions.
- Collaborated on AI product development and research initiatives focused on automation and intelligent workflows.

Technologies:

Python, OpenCV, TensorFlow, PyTorch, OCR, NLP, Machine Learning, Computer Vision.

testneo

AI Engineer

November 2024 - February 2025 (4 months)

India

Worked on AI-powered automation and developer productivity solutions focused on intelligent API testing and document-driven software validation.

Key Contributions:

- Developed an AI-powered API Test Case Generator capable of generating functional, security, and edge-case test scenarios from API documentation and business requirements.
- Built document intelligence pipelines supporting PDF, DOCX, OpenAPI (JSON/YAML), and TXT file processing.
- Implemented Retrieval-Augmented Generation (RAG) workflows using ChromaDB and semantic search for context-aware test generation.
- Integrated Groq LLMs to automate test case creation, validation workflows, and intelligent software testing assistance.
- Designed export pipelines supporting JSON, CSV, Markdown, and Python (pytest) formats.

Technologies:

Python, Streamlit, Groq, ChromaDB, LangChain, FastAPI, Vector Search, RAG, OpenAPI, Generative AI.

Illinois Institute of Technology

Data Analyst

May 2023 - July 2023 (3 months)

Enhanced data integrity for over 50,000 records, increasing accuracy by 15% through efficient cleaning techniques.

Developed interactive dashboards, reducing reporting time by 30% and improving data visualization clarity.

Proposed process improvements that increased data processing efficiency by 20%.

Education

Dayalbagh Educational Institute

Master of Technology, Computational Science · (September 2022 - June 2024)

Dayalbagh Educational Institute

Postgraduate Degree, Data Processing and Data Processing Technology/
Technician · (September 2021 - August 2022)

MJPRU, Bareilly

Bachelor of Science, PCM · (August 2017 - October 2020)

Government Industrial Training Institute - Bijnor

ITI COPA, Computer Science · (October 2020 - August 2021)